

MCA SEMESTER-I
School of Computer and Systems Sciences,
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Computer Fundamental (CS-402)

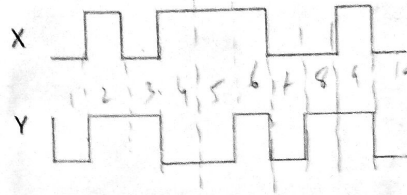
Date: 02.03.2023

Total Marks: 50

Duration: 3hr

All Compulsory

1. Perform the a) OR b) AND c) EX-OR d) EX-NOR e) NOT for following pulse waveforms: 5



2. Prove the theorems: a) $x+x=x$ b) $x.x=x$ c) $x+1=1$ d) $x+xy=x$ 5

3. Simplify the Boolean Function F using don't care condition 'd' : 5

$$F = A'B'D' + A'CD + A'BC$$

$$d = A'BC'D + ACD + AB'D'$$

Implement the simplified function using 2-input gates.

4. Implement a 3x8 decoder. 5

5. Implement a full adder using 2-input gates. 5

6. Implement the following function with a 8X1 multiplexer: 5
 $F(A,B,C,D) = \Sigma(0,1,3,4,8,9,15)$

7. Implement a 3-bit magnitude comparator. 7

8. Implement a 8x4 ROM for the following: 7

x	y	z	F1	F2	F3	F4
0	0	0	0	1	1	0
0	0	1	0	1	0	0
0	1	0	0	0	1	0
0	1	1	0	0	0	1
1	0	0	1	1	0	0
1	0	1	1	1	1	0
1	1	0	1	0	0	0
1	1	1	1	0	0	1

9. Implement the following functions with PLA: 6
 $F_1(A,B,C) = \Sigma(3,5,6,7)$
 $F_2(A,B,C) = \Sigma(0,2,4,7)$